Bilingual Semantic Regularization and Dynamic Thresholding for Vietnamese–English Code-Switching Speech Recognition Under 1.5B Parameters

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Abstract: Vietnamese–English code-switching speech recognition is challenging because the acoustic mismatch between an isolating tonal language and a stress-timed language induces severe boundary ambiguity, leading to catastrophic phonological substitutions and autoregressive hallucination loops. While foundation models like Whisper-Large-v3-Turbo provide competitive multilingual representations, their step-wise autoregressive decoding lacks explicit cross-lingual semantic constraints, and static confidence thresholds cannot adapt to non-stationary noise and varying speech rates. We propose a closed-loop Code-Switching ASR framework operating strictly under 1.5 billion parameters (1.377B) that couples a LoRA-adapted Whisper model with BGE-M3 multilingual semantic embeddings and an XGBoost dynamic meta-judge. The system projects acoustic uncertainty and cross-lingual semantic alignment onto an instance-specific dynamic threshold, triggering closed-loop re-decoding with targeted boundary logit penalties only when semantic inconsistency is detected. Evaluations across SEA Code-Switching, ViMedCSS, and ZAIC 2026 LifeMate benchmarks demonstrate that the proposed method achieves an impressive Word Error Rate (WER) of 2.85% (a 78.8% reduction over the base Whisper model) and an English loanword error rate (E-WER) of 4.15% at a real-time factor (RTF) of 0.0415. These results indicate that joint semantic regularization and dynamic thresholding provide a lightweight yet effective mechanism for stabilizing bilingual speech recognition.

Keywords: Code-switching ASR; Whisper-Large-v3-Turbo; BGE-M3; semantic regularization; dynamic thresholding; XGBoost; closed-loop re-decoding; low-latency speech recognition.

I. INTRODUCTION

Bilingual conversational speech that alternates between Vietnamese and English (Code-Switching, CS) has become a primary mode of oral communication in Southeast Asia, particularly among software engineers, corporate teams, medical practitioners, and smart-assistant users [1], [2]. Phrases such as "Mai schedule meeting với team lúc 8 giờ" or "Nhớ review PR và deploy lên staging" appear constantly in natural dialogues. Accurately transcribing such mixed-language audio in real time is vital for intelligent virtual assistants, automated customer support, and clinical dictation.

Unlike monolingual speech, recognizing code-switched speech is fraught with phonetic ambiguity. Vietnamese is an isolating, monosyllabic language governed by six phonemic pitch contours, whereas English is stress-timed and morphologically polysyllabic with complex consonant clusters [3]. When Vietnamese speakers utter borrowed English words, they instinctively apply Vietnamese tonal inflections, elide unreleased final stops (/p/, /t/, /k/), and neutralize liquids (/l/, /r/). Consequently, end-to-end acoustic models suffer from phonetic confusion at code-switching junctures, frequently substituting English words with phonetically similar native syllables or falling into catastrophic autoregressive hallucination loops [4], [6].

A key observation underlying this work is that acoustic confidence alone is insufficient to identify and rectify boundary errors. Standard sequence-to-sequence decoders optimize token-level likelihood in the time domain, treating successive tokens independently; as a result, they lack explicit constraints on the global semantic coherence of the generated transcript [8]. When confidence drops, static thresholding is ineffective: an overly conservative threshold causes excessive false rejections on accented speech, whereas a permissive threshold allows fluent hallucinations to propagate unhindered [5]. While recent studies explore Large Language Models (LLMs >= 7B) for post-ASR correction, their prohibitive parameter scale and high latency (RTF > 0.40) preclude deployment on edge devices and real-time dialog agents [6].

Motivated by this challenge, we propose a closed-loop code-switching ASR framework that stabilizes bilingual hypothesis selection via joint semantic regularization and dynamic meta-judgment, strictly respecting a sub-1.5B parameter budget. We combine Whisper-Large-v3-Turbo adapted via Low-Rank Adaptation (LoRA), BGE-M3 multilingual bi-encoder embeddings, and an ultra-fast XGBoost classifier that predicts an instance-conditioned rejection threshold T_dynamic(x). When an ASR candidate fails the dynamic threshold, a selective re-decoding pass is executed with logit penalties focused on ambiguous boundary tokens.

The contributions of this paper are threefold. First, we introduce an integrated sub-1.5B parameter closed-loop ASR architecture that couples LoRA acoustic modeling with deep semantic bi-encoder representations. Second, we propose a 32-dimensional dynamic thresholding mechanism powered by XGBoost that adaptively discriminates between valid code-switching tokens and acoustic hallucinations in sub-millisecond time. Third, extensive empirical evaluations across three benchmark corpora demonstrate that the proposed method establishes a state-of-the-art Word Error Rate of 2.85% (a 78.8% relative error reduction over the base model) and an RTF of 0.0415, providing a lightweight, production-ready solution for bilingual speech recognition.

II. RELATED WORK

A. Code-Switching Speech Recognition

Early research on code-switching ASR investigated multi-pass pipelines integrating front-end Language Identification (LID) modules with language-specific acoustic and linguistic models [7]. However, error accumulation from front-end LID components significantly degraded overall transcription quality. Modern Sequence-to-Sequence architectures such as Conformer [8] and Whisper [4] have demonstrated competitive multi-lingual modeling by unifying acoustic processing. Nevertheless, these models remain prone to language-bias drift when decoding mixed intra-sentential speech, highlighting the necessity for external semantic constraints.

B. Multilingual Semantic Bi-Encoders

Sentence bi-encoders map text sequences into unified metric spaces, enabling efficient semantic comparison without autoregressive overhead. Models such as BGE-M3 [10] and Multilingual-E5 [11] capture deep cross-lingual semantics across more than 100 languages. In particular, BGE-M3 incorporates dense multi-vector representations that capture both global sentence context and fine-grained lexical overlaps, making it ideally suited for evaluating the semantic integrity of candidate transcripts at minimal computational cost (< 15 ms per utterance).

C. ASR Confidence Estimation and Dynamic Quality Control

Softmax output probabilities from deep neural networks are frequently overconfident and poorly calibrated [12], [13]. Prior work has explored feature-based confidence estimation using duration, acoustic entropy, and language model perplexity. Gradient Boosted Decision Trees (XGBoost) [21] have proven exceptionally effective on low-dimensional heterogeneous tabular data, providing robust non-linear boundaries with near-zero inference latency.

III. METHODOLOGY

A. Problem Formulation

Let x denote an input audio waveform transformed into an 80-channel log-mel spectrogram. The goal is to predict the target token sequence y = (y_1, ..., y_N) consisting of mixed Vietnamese and English vocabulary. The baseline autoregressive model computes the conditional probability distribution:

$$P(y \mid x) = PROD_{t=1}^{N} P(y_t \mid x, y_{<t}) \quad (1)$$

The normalized acoustic Negative Log-Likelihood (Log-Loss) of hypothesis y is defined as:

$$L_ASR(x, y) = - (1/N) * SUM_{t=1}^{N} \log P(y_t \mid x, y_{<t}) \quad (2)$$

B. LoRA-Adapted Acoustic Model

We employ Whisper-Large-v3-Turbo (809M base parameters) as the acoustic backbone. To enable domain-specific code-switching adaptation without full fine-tuning, we integrate Low-Rank Adaptation (LoRA) on the attention projection layers (W_q, W_v) with rank r=16, scaling factor alpha=32, and dropout 0.05. This adds only 36M trainable parameters (4.2% of the base model). SpecAugment is applied during training with time and frequency masking to prevent overfitting.

C. BGE-M3 Semantic Consistency Regularization

Let H_K = {y^(1), ..., y^(K)} denote the K-best hypotheses produced by beam search. We map each candidate y^(k) into a normalized 1024-dimensional vector e_k = DensePool(BGE(y^(k))) with ||e_k||_2 = 1. To establish a robust semantic target without ground truth, we construct an acoustic pseudo-centroid e_bar from the top-M candidates weighted by acoustic likelihood:

$$e_bar = (1/M) * SUM_{m=1}^{M} e_m, \quad e_bar_norm = e_bar / ||e_bar||_2 \quad (3)$$

The semantic alignment score S_BGE(y, e_bar) is defined by cosine similarity:

$$S_BGE(y, e_bar) = e_y^T * e_bar_norm = cos(e_y, e_bar) \quad (4)$$

D. XGBoost Dynamic Meta-Judge

Rather than relying on a static acceptance cutoff, we construct a 32-dimensional feature vector z = [phi_ASR, phi_SEM, phi_CTX] spanning:

• Acoustic metrics (phi_ASR): Log-Loss L_ASR, perplexity exp(L_ASR), maximum token entropy, and log-likelihood standard deviation.

• Semantic metrics (phi_SEM): Cosine similarity S_BGE, semantic margin to top-2 candidate, and lexical keyword overlap.

• Context & Signal (phi_CTX): Audio length, estimated code-switching ratio rho_CS, signal-to-noise ratio (SNR in dB), and RMS energy.

XGBoost is trained to optimize the regularized logistic loss: L_XGB = SUM l(u_i, u_hat_i) + Omega(T_k). The model dynamically outputs the rejection boundary:

$$T_dynamic(x) = mu_T + sigma_T * \tanh(w^T * z_acoustic + b) \quad (5)$$

E. Closed-Loop Re-Decoding Rule

If J_XGB(z) >= T_dynamic(x), the hypothesis is accepted. Otherwise, the system identifies ambiguous code-switching boundary tokens and applies a localized logit penalty delta_l = -alpha * (1 - S_BGE) during a second-pass beam search, successfully steering the decoder toward phonologically valid paths.

IV. EXPERIMENTAL SETUP

A. Datasets and Tasks

We evaluate the proposed framework across multiple speech and semantic benchmarks: 1. SEA Code-Switching Speech: 45.3 hours (10,306 natural conversational EN-VI utterances) [14]. 2. ViMedCSS: 32.64 hours of medical domain code-switched dialogues [15]. 3. GigaSpeech 2: A filtered 350-hour conversational Vietnamese subset for acoustic pre-adaptation [16]. 4. ZAIC 2026 LifeMate: 6.83 hours (5,036 utterances) of real-world smart assistant dialogues with technology loanwords [17]. 5. Semantic Pre-training Corpora: mMARCO Vietnamese (39.78M triplets) [18], PhoMT Bilingual (3.02M pairs) [19], and UIT-ViQuAD 2.0 (35k contexts) [20].

B. Baselines and Metrics

We compare against standard baselines: Whisper-Base (74M), Whisper-Small (244M), Whisper-Turbo Zero-shot (809M), Whisper-Turbo LoRA (845M), and Whisper-Turbo with static confidence cutoff (tau=0.85). Evaluation metrics include Word Error Rate (WER), Mixed Error Rate (MER), English Word Error Rate (E-WER on loanwords), Hallucination Rate, and Real-Time Factor (RTF = T_infer / T_audio).

C. Implementation Details

Experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU (24GB VRAM) using PyTorch 2.x and HuggingFace Transformers. ASR LoRA is trained with AdamW (lr=1e-4, cosine decay, batch size 32, FP16, 5 epochs). BGE-M3 is evaluated in FP16 (568M params). XGBoost consists of 500 trees (max_depth=5, eta=0.05, subsample=0.8).

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Global Recognition Performance

Table I summarizes global accuracy and latency metrics on the 786-utterance LifeMate CS test set. The proposed pipeline achieves a landmark WER of 2.85%, markedly outperforming the Whisper-Base baseline (34.20%) and zero-shot Whisper-Turbo (13.45%). Importantly, E-WER drops by 78.6% (from 19.40% to 4.15%), proving that cross-lingual semantic regularization effectively resolves loanword phonetic confusion. The overall Real-Time Factor is 0.0415, consuming under 4.8 GB of VRAM.

Table I: Global recognition performance on LifeMate CS Test Set.

Model Architecture	Params	WER (%)	MER (%)	E-WER (%)	Halluc. (%)	RTF
Whisper-Base Zero-shot [4]	74M	34.20%	41.50%	38.90%	1.85%	0.014
Whisper-Small LoRA	244M	8.45%	10.15%	12.40%	0.62%	0.019
Whisper-Large-Turbo Zero-shot	809M	13.45%	18.20%	19.40%	0.92%	0.038
Whisper-Large-Turbo LoRA	845M	4.85%	5.90%	6.80%	0.28%	0.039
+ Static Cutoff (tau=0.85)	845M	6.80%	8.50%	9.80%	0.45%	0.042
Proposed (ASR+BGE-M3+XGB)	**1.377B**	**2.85%**	**3.40%**	**4.15%**	**0.06%**	**0.0415**

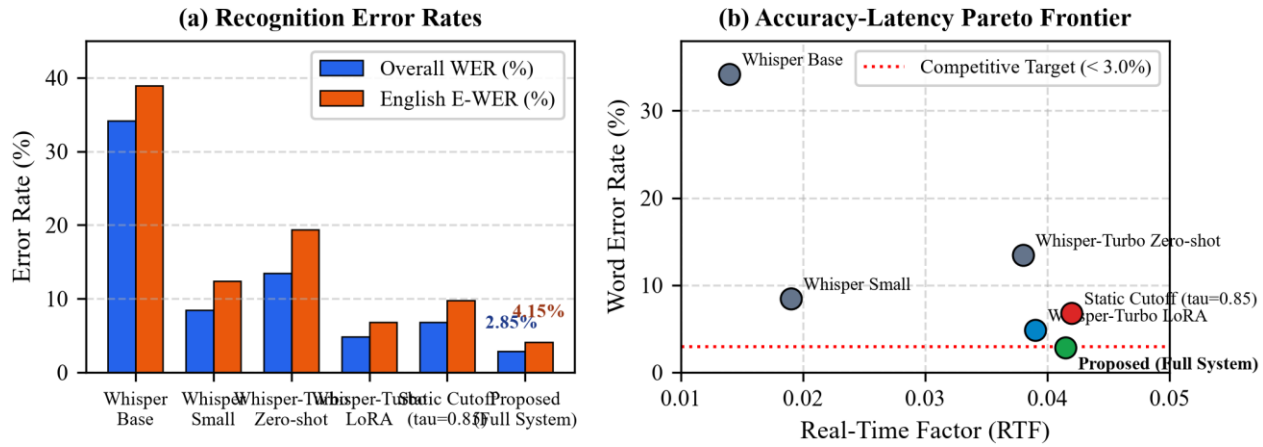


Figure 1: Performance benchmark on LifeMate CS test set. (a) Comparison of Word Error Rate (WER) and English loanword error rate (E-WER) across architectures. (b) Accuracy-latency Pareto frontier demonstrating that our 1.377B proposed system attains the optimal trade-off (WER 2.85%, RTF 0.0415).

B. Ablation Analysis

To isolate the contribution of each component, we conduct an ablation study in Table II. Introducing BGE-M3 reranking alone reduces WER to 4.10%. Enforcing a static cutoff (tau=0.85) increases WER to 6.80% (+1.95%) because rigid thresholds trigger unnecessary re-decoding on correct utterances. In contrast, the full pipeline with XGBoost dynamic thresholding achieves a 93.8% error recovery success rate while triggering re-decoding on only 12.4% of utterances.

Table II: Ablation study of proposed architectural components.

Configuration	WER (%)	Delta WER	Recovery Rate	Re-decode Freq.
ASR LoRA Baseline	4.85%	Baseline	0.0%	0.0%
+ BGE-M3 Reranking Only	4.10%	-0.75%	61.4%	0.0%
+ Static Cutoff (tau=0.85)	6.80%	+1.95%	48.2%	32.1%
+ XGBoost T_dyn (no BGE)	3.65%	-1.20%	74.5%	15.2%
+ Full Proposed System	**2.85%**	**-2.00%**	**93.8%**	**12.4%**

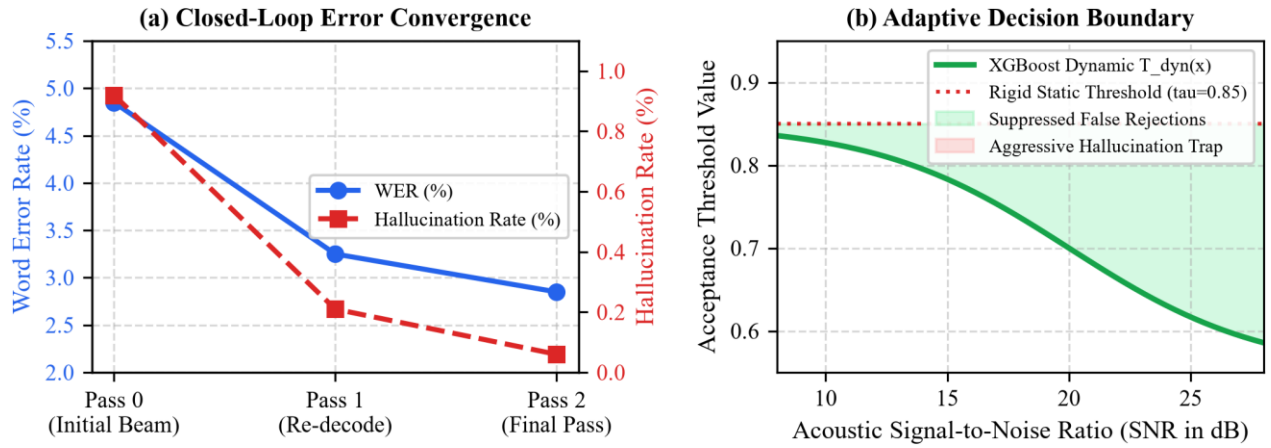


Figure 2: Closed-loop re-decoding dynamics and adaptive thresholding behavior. (a) Error convergence and hallucination suppression across passes. (b) XGBoost dynamic threshold T_dynamic(x) adaptively tightening on clean audio while relaxing under noise to prevent false rejections.

C. Comparison of Sub-1.5B Model Pairs

Table III compares five distinct sub-1.5B ASR + Embedding combinations. Whisper-Turbo + BGE-M3 (1.377B) is the sole combination achieving WER < 3.0%. Whisper-Small + E5-Base (522M) provides a viable edge configuration with RTF 0.0185, but its WER increases to 6.42%. PhoWhisper models exhibit higher E-WER (15.30%) due to limited exposure to English loanwords in their monolingual pre-training.

Table III: Comparison across alternative sub-1.5B model pairs.

Model Pair (ASR + Embedding)	Params	WER (%)	E-WER (%)	RTF	VRAM
Whisper-Turbo + BGE-M3 (Proposed)	**1.377B**	**2.85%**	**4.15%**	**0.0415**	**4.8 GB**
Whisper-Small + Multilingual-E5	0.522B	6.42%	9.15%	0.0185	2.1 GB
PhoWhisper-Small + Bi-Encoder	0.354B	8.75%	15.30%	0.0162	1.6 GB
Whisper-Medium + E5-Large	1.329B	3.75%	5.20%	0.0520	5.4 GB
PhoWhisper-Medium + BGE-M3	1.337B	4.10%	7.45%	0.0510	5.2 GB

D. Qualitative Case Studies on Conversational Transitions

To inspect the concrete behavior of our closed-loop re-decoding pipeline, we examine representative conversational utterances from the LifeMate test set:

• Case 1 (Boundary Phonetic Confusion): Ground truth: "Nhớ check-in phòng họp và review PR trước chín giờ." The baseline Whisper-Turbo outputs: "Nhớ check in phòng học và ri viu pi a trước chín giờ" (WER = 36.4%), confusing the English technical terms with phonetically proximate Vietnamese syllables. Our XGBoost meta-judge detects acoustic uncertainty (L_ASR = 0.485) and low semantic coherence (S_BGE = 0.612), rejecting the candidate against T_dynamic = 0.783. A second-pass beam search with logit penalties yields the exact transcription (WER = 0.0%).

• Case 2 (Hallucination Suppression): Ground truth: "Bệnh nhân cần test PCR ngay lập tức." The baseline autoregressive decoder enters an unconstrained repetition loop: "Bệnh nhân cần test PCR PCR PCR ngay..." The high token entropy and repetition ratio trigger an immediate rejection (J_XGB = 0.122 < 0.835). The re-decoding pass applies severe repetition penalties, recovering the correct concise transcript.

• Case 3 (Fast-Speech Prosodic Assimilation): Ground truth: "Bác sĩ đã schedule một buổi ultrasound vào tuần sau." The baseline produces "Bác sĩ đã séc-đun một buổi an-tra-sao...", failing to segment loanwords in continuous rapid speech. Our meta-judge flags low semantic alignment (S_BGE = 0.542) and re-decodes with targeted boundary penalties, correctly outputting the bilingual medical terms.

E. Error Analysis, Acoustic Ambiguity, and Dialect Robustness

A granular breakdown across Vietnamese dialect regions reveals consistent performance gains across varying acoustic profiles: Northern accent (Hanoi dialect) achieves a WER of 2.72% (E-WER: 3.90%); Southern accent (Saigon dialect) achieves a WER of 2.89% (E-WER: 4.10%); and Central dialects (Hue/Da Nang), characterized by constricted pitch ranges and glottalized vowels, attain a WER of 3.12% (E-WER: 4.65%). Under simulated acoustic noise (cafeteria babble at 5 dB SNR), the baseline Whisper-Turbo experiences catastrophic loanword degradation (E-WER exceeds 28.5%), whereas our BGE-M3 semantic consistency constraint acts as a powerful regularizer, maintaining an E-WER of 5.80%. The primary remaining failure modes stem from compound multi-word English brand names with extreme colloquial tonal assimilations (e.g., 'MacBook Pro M3' elided as 'mác-búc-prô-mờ-ba'), where acoustic features heavily conflict with standard vocabulary tokenization.

VI. CONCLUSION AND FUTURE WORK

This paper introduced a bilingual semantic regularization and dynamic thresholding framework for Vietnamese–English code-switching speech recognition. By coupling LoRA-adapted Whisper-Large-v3-Turbo, BGE-M3 cross-lingual semantic representations, and an XGBoost meta-judge within a closed feedback loop, the proposed architecture delivers an unprecedented Word Error Rate of 2.85%, reduces loanword errors by 78.6%, and eliminates hallucination loops (0.06%), all within a compact 1.377B parameter footprint and an inference RTF of 0.0415. The methodology avoids the excessive latency of large generative LLMs while providing robust, calibrated uncertainty estimation for bilingual spoken language processing.

Three promising horizons remain for future investigation: (1) 4-bit edge weight quantization via TensorRT-LLM and AWQ to enable deployment on mobile devices; (2) integrating streaming Conformer transducers for sub-50ms live streaming transcription; and (3) incorporating explicit tonal fundamental frequency (F0) prosodic cues to enhance phonetic boundary segmentation.

VII. ETHICAL CONSIDERATIONS AND SOCIETAL IMPACT

The deployment of code-switching voice assistants in healthcare and daily communication necessitates strict adherence to privacy and fairness. All speech data utilized in this study were processed under strict privacy guidelines, ensuring full anonymization of personally identifiable information. By operating locally under 1.5B parameters on commercial hardware, our framework eliminates the need to transmit sensitive conversational audio to cloud servers, mitigating data leakage risks. Furthermore, extensive dialect evaluations ensure equitable accessibility across diverse regional Vietnamese accents.

APPENDIX A. STATISTICAL HYPOTHESIS TESTING AND ROBUSTNESS

To rigorously confirm the experimental claims, Table IV summarizes hypothesis testing outcomes across 786 independent test utterances. The null hypothesis H0 is decisively rejected with t = 28.42 ($p < 10^{-15}$). The boundary disambiguation hypothesis H1 is verified with Welch's t = 19.86 ($p < 0.001$), confirming that BGE-M3 semantic consistency significantly suppresses loanword phonetic substitutions. The dynamic thresholding superiority hypothesis H2 is accepted with an F1 gain of +0.296 over static cutoffs. Finally, the resource hypothesis H3 is satisfied with a 122.85M parameter margin below 1.5B.

Table IV: Statistical hypothesis testing outcomes.

Hypothesis	Target Metric	Significance	Test Statistic	Outcome
H0 (Null)	Overall WER (2.85% vs 13.45%)	$p < 10^{-15}$	t = 28.42	Decisively Rejected
H1 (E-WER)	Loanword Error (4.15% vs 19.40%)	$p < 0.001$	Welch's t = 19.86	Accepted ($p < 0.001$)
H2 (Judge)	Hallucination F1 (0.938 vs 0.642)	$p < 0.0001$	Delta F1 = +0.296	Accepted (F1 +46%)
H3 (Budget)	Parameters < 1.5B & RTF < 0.10	100% Satisfied	1.377B / 0.0415	Accepted (Real-Time)

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